

HCI and Usability



What is HCI?

Human-Computer Interaction (HCI)

ACM definition (Association for Computing Machinery)

Human-Computer interaction is a discipline concerned with the design, evaluation and implementation of interactive computing systems for human use and with the study of the major phenomena surrounding them

HCI is very important

Neglecting the aspects concerning human-computer interaction means neglecting the user (= consumer)

HCI studies are mainly focused on the *interface*

The communication "gateway" between the user and the computer

Usability

HCI branch that studies methods to evaluate and improve computer "products"

Unfortunately, *usability tests* are sometimes carried out only when the product has been fully created...

ISO definition (ISO = International Organization for Standardization)

The extent to which a product can be used by specified users to achieve specific goals with effectiveness, efficiency, and satisfaction in a specified context of use

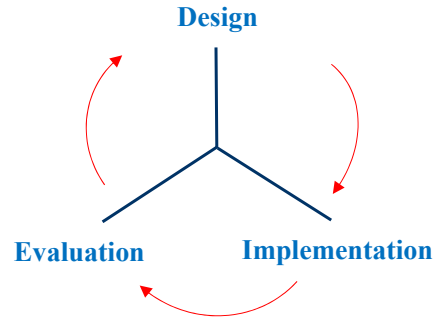
Usability Engineering

When the evaluation process is integrated in the production cycle (*design for usability*)

Usability Engineering

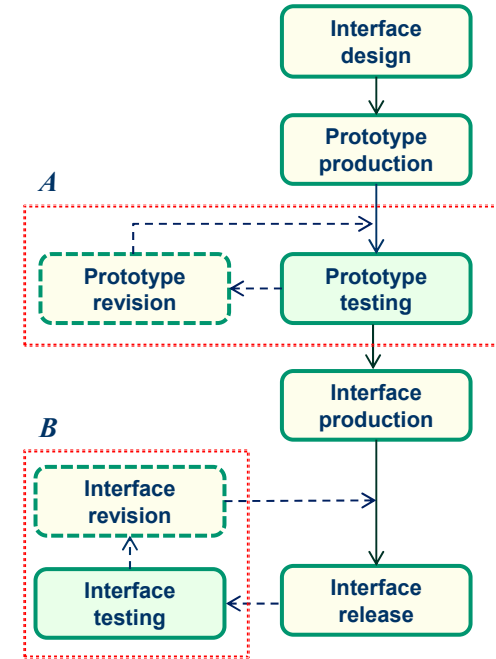
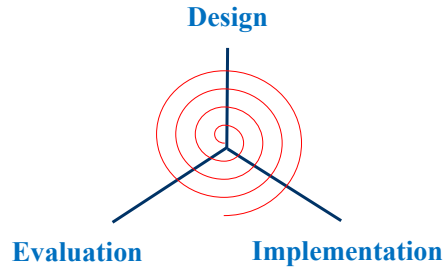
Iterative design

(bad approach:
each iteration
corresponds to a
new release)



Spiral model

(good approach:
the production
process includes
several iterations)



B is usually much more expensive than A

Web Usability

Rules to "properly" design websites

So that they can be easily accessed, surfed, read, etc.

If a website is not usable, the user may leave it

Users want to think as little as possible... and want to achieve their goals quickly!

The usability of a website depends on its *information architecture*, navigation system, graphic layout, etc.

Information Architecture: how is the *information* arranged among pages, and how is it organized within each page?

Usability testing methods...

- Based on experts
 - Cognitive walkthrough
 - Detailed analysis of sequences of actions and/or commands to verify the "ease of learning" of the interface
 - Heuristic evaluation
 - Once some "usability criteria" have been defined, the product is examined to check whether all such criteria are satisfied
- Based on users
 - Observation
 - a) *Direct observation*: the user is observed while doing something; b) *Video-recording*: the user is video-recorded while doing something; c) *Log files analysis*: the "dialog" between user and computer is reconstructed from log data (e.g., web server's log files); d) *Verbal protocols*: what the user says is recorded and then analyzed
 - Questionnaires
 - The user expresses his or her "opinion" by answering specific questions
 - Experimental evaluation
 - Implies a "scientific" analysis of some relevant factors

The "science" of Usability

... usability testing methods

A "sophisticated" experimental method: eye tracking



Some examples of "usability rules"

Web Usability (according to Jakob Nielsen)

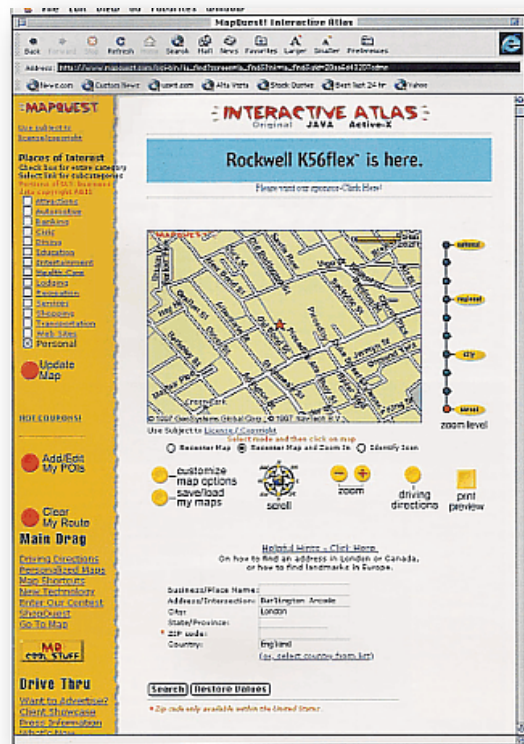
Page design



Best exploitation of the screen

Pages should be mainly occupied by relevant content

- Empirical rule: the "useful" content should occupy at least 50% of the page (and be as close as possible to 80%)
- To gain space, try removing interface elements (one at a time): if the page still "works", then those elements are not necessary...



Multiplatform design

It is not possible to foresee the user's screen size

- The content should automatically adapt to the browser's window: liquid layouts are generally preferable to fixed-width layouts
- Enlarging the font size should not cause any problem in page display

Avoid using non-standard content

Not all users will be able to exploit new technologies, and not all users will be willing to install a plugin

Quick responses

If the user has to wait a long time to access a website, they may look for alternatives

Even before seeing the home page...

Avoid exceeding 10 seconds

In general: limit for the user to perceive a response as "instantaneous" → 1/10 sec;
limit for the "flow of thoughts" not to be interrupted → 1 sec; limit for the user not to
distract his or her attention → 10 sec

Useful information in the homepage should arrive very rapidly

Details may take more time...

Links

The meaning of links should be as clear as possible

Avoid cryptic words, phrases or pictures

Add titles

Indicate the destination site and/or section, as well as details about the kind of information that will be found

E.g. In the same period, `<a href="gv.html" title="Biography of Giuseppe Verdi">Giuseppe Verdi</a>` began...

In the same period, [Giuseppe Verdi](#) began...



Biography of Giuseppe Verdi

Avoid using non-standard colors/styles

The user is "accustomed" to underlined words and to certain colors for links... in any case, always be **consistent!**

Frames

May be misleading for the user

The invention of the Web has concentrated in the single idea of *page* different concepts:

a) Visual information unit; b) Navigation unit; c) Address (URL)

Main problems:

- The URL shown in the browser's address bar does not identify univocally the displayed content
- Bookmarks may not identify the specific content displayed at a certain moment
- Wrong page portions may be printed if the correct frame is not properly selected
- Pages with frames have accessibility problems!

Web Usability (according to Jakob Nielsen)

Writing for the Web...

Text should be concise and readable

- Reading on a screen is more uncomfortable than reading printed text
- Bulleted lists can help to logically subdivide the text into main concepts
- Important words should be emphasized
- Each page should not contain much information
- Headings should be highly significant (users do not like wasting time reading what they are not interested in)

... writing for the Web

Headings should possibly be chosen so that:

- They clearly explain the topic of the page
- The first word is the most important (for them to be easily identifiable in alphabetical lists)
- They start with different words (for them to be easily distinguishable)

For high readability:

- Text and background colors should be well contrasted
- Only solid colors or subtle patterns should be used for the background
- The font size should not be too small

Multimedia...

Main problem: response times

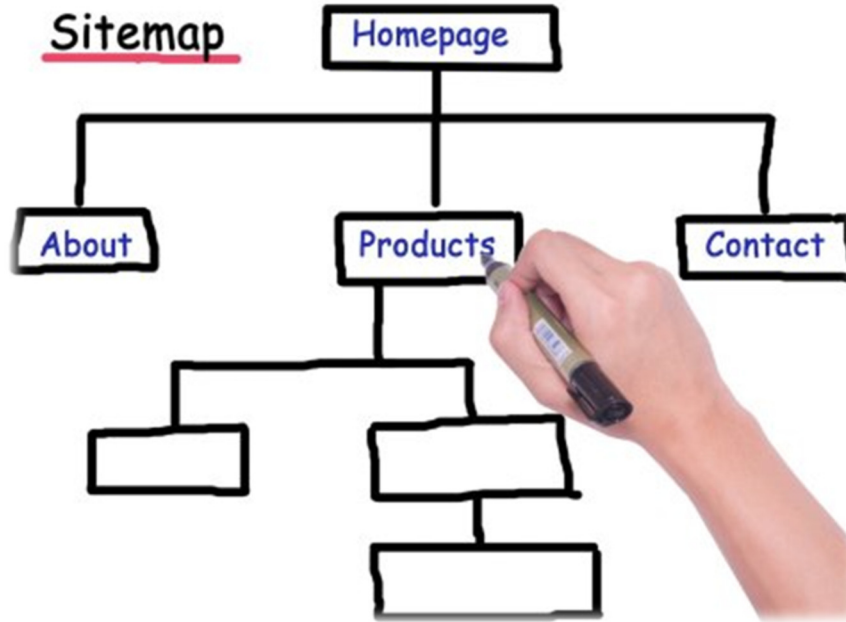
Multimedia files are generally big

Images, photos, animations

- Should be used only when necessary
 - "General" pages (e.g., the home page and pages directly reachable from it) should contain a limited number of multimedia elements (initially, the user hasn't expressed a clear interest in the website yet)
- Should be limited in size
 - Including only important elements

Web Usability (according to Jakob Nielsen)

Website design



Home page

Should be designed with **special care**

The home page should **clearly** communicate what the website is about

Should provide three basic functions

1) Directory of main contents (navigation links), 2) Summary of latest changes, and 3)
Possible internal search engine

Avoid initial "splash pages"

Apart from those that warn the user about the kind of content of the website (e.g.,
adult websites)

Navigation...

A navigation interface should help the user find the answers to three basic questions:

- Where am I now?
- Where have I been?
- Where can I go?

Where am I now?

The current position should be provided with reference to both the Web and the website

- The website's name (and/or logo) should be repeated in all pages, always in the same location
- The position within the website should be indicated through a representation of its structure (e.g., using "breadcrumbs")
- Titles of pages (<title> tag) should be meaningful

... navigation

Where have I been?

- The Web is memoryless, and the website cannot "know" which pages have already been visited
 - Unless *cookies* are exploited
- The browser's predefined functions can be of help ('Back' and 'Next' buttons, history), as well as links' colors (different for already visited and not yet visited links)

Where can I go?

- Is expressed by links in the page (although it is not possible to indicate all destination pages)
- A good information architecture is necessary